survive for 1.5 days inside a cell but only about 6 hours outside a cell. Routes of HIV transmission include unsafe sexual contact, blood-contaminated needles, organ transplants, blood transfusion and vertical transmission from HIV infected mother to child. HIV is not transmitted by insects or by casual contact.

After getting into the body of the person, the virus enters into macrophages where **RNA genome** of the virus replicates to form **viral DNA** with the help of the enzyme **reverse transcriptase**. This viral DNA gets incorporated into the DNA of host cells and directs the infected cells to produce viral particles. The macrophages continue to produce virus and in this way acts like a HIV factory. Simultaneously, HIV enters into helper T-lymphocytes, replicates and produces progeny viruses. The progeny viruses released in the blood attack other helper T-lymphocytes. This is repeated, leading to a progressive decrease in the number of helper T lymphocytes in the body of the infected person. During this period, the person suffers from bouts of fever, diarrhoea and weight loss. Due to decrease in the number of helper T lymphocytes, the person starts suffering from infections and becomes immune deficient and unable to protect against any infection.

A simple blood test is available that can determine whether the person has been infected with HIV. The **ELISA** test (Enzyme Linked ImmunoSorbent Assay) detects the presence of HIV antibodies. It is a preliminary test. **Western blot** test is more reliable and a confirmatory test. It detects the viral core proteins. If both tests detect the presence of the antibodies, the person is considered to be HIV positive.

AIDS has no cure. **Prevention** of AIDS is the best option. Advocating safe sex and promoting regular check-up, safe blood for transfusion, use of disposable needles, use of condoms during sexual contact, prevention of drug abuse, AIDS awareness programme by **NACO** (National AIDS Control Organisation), **NGO**s (Non-Governmental Organisations) and **WHO** are to prevent the spreading of AIDS.

7.5 Autoimmune diseases

Autoimmunity is due to an abnormal immune response in which the immune system fails to properly distinguish between self and non-self and attacks its own body. Our body produces antibodies **(auto antibodies)** and **cytotoxic T cells** that destroy our own tissues. If a disease-state results, it is referred to as auto-immune disease. Thus, autoimmunity is a misdirected immune response. Autoimmunity is evidenced by the presence of **auto antibodies and T cells** that are reactive with host antigens. When the cells act as antigens in the same body, they are called autoantigens.

Autoimmune diseases in human can be divided into two broad categories, namely organ-specific and non-organ-specific (systemic) autoimmune diseases. In organ-specific disease, the autoimmune process is directed mostly against one organ. The autoantibodies may block the functions performed by the organs. Examples include Hashimoto's thyroiditis, Graves' disease (thyroid gland) and Addison's disease (adrenal glands).

In non-organ specific (systemic) disorders, autoimmune activity is widely spread throughout the body. Rheumatoid arthritis and multiple sclerosis are example for systemic disorder.

7.5.1 Tumour immunology

A **tumour** or **neoplasm** is a group of cells whose growth has gone unchecked. When a tumour continues to grow and invades healthy tissue, it is called cancer. They spread to other parts of the body from the tumour and give rise to secondary tumour. This is known as **metastasis**. Tumour may be benign or malignant depending on its characteristics. Benign or non-cancerous tissues are capable of indefinite growth and do

XII Std Biology-Zoology Chapter-7 EM.indd 128 23/12/2022 12:39:59